

Cozmo technical report

Handheld phone capture to a dimensioned floor plan with damage annotations.

Every number below comes from a file in this repository and names the script that rebuilds it. Anything not yet measured is marked PENDING with what it waits on. `scripts/regenerate_all.sh` rebuilds all of it.

1. Architecture

One contract, one backend, three front ends.

The contract first. `cozmo/contracts/models.py` defines the plan. No

dimension is ever a bare number: every `*_m`, `*_m2` and `*_deg` field is a

Measurement with value, `ci_low`, `ci_high`, unit, method and

`ci_level`. A test walks emitted JSON and fails on any bare number.

Referential integrity is checked at load, so a plan that parses has a

consistent graph and the evaluator never repairs input before scoring it. The

schema is generated from the models, so the two cannot drift.

The shared backend. Tiers differ only in how they make geometry. Once a

tier has a point cloud with poses, the rest is shared: `cozmo/lidar/levels.py`

(floor and ceiling), `walls.py` (Manhattan frame, wall faces), `rooms.py` and

`cells.py` (rooms), `openings.py` (doors, adjacency), `uncertainty.py`

(intervals), `render/plan.py`. Damage is separate and tier-agnostic.

Provenance. `run` writes `plan.json`, `plan.png`, `run_manifest.json`,

`drift_report.json` and `debug/`. The manifest holds a SHA-256 of every input

file the tier may read, the resolved config and its hash, the git commit, the

seed, stage timings and library versions. The plan holds no wall-clock field,

so the same input, config and seed give a byte-identical `plan.json`.

Tier isolation is enforced in code, not by convention. Every file access

goes through `StrayScan.open`, which raises on a file the tier may not read:

the video tier sees only `rgb.mp4`, the photo tier is refused a scan folder

outright because `depth/` holds png files that would otherwise pass as room

photos. The run manifest hashes only the permitted files, so provenance never

touches data the tier could not have used (`tests/test_stray.py`).

2. Tiers and devices

tier	input	reconstruction	median wall error against tape
lidar	one Stray Scanner scan for the whole property	classical geometry, no trained model	no tape ground truth exists
photo	2 to 8 stills per room	MapAnything poses, camera-height scale, same backend	1.4% , worst 21.5%
video	one clip per room	COLMAP SfM in chunks, camera-height scale, same backend	32.9% , worst 137.4%

Wall errors from `docs/benchmark/eval.json`, rebuilt by

`scripts/benchmark_all.py`, over 12 photo walls and 16 video walls with tape

readings. The photo tier is an order of magnitude better than the video tier

on the same rooms with the same scale cue and the same room fitter; section 6

explains why.

Stray Scanner format. `rgb.mp4` 1920x1440 HEVC 60 fps stored rotated;

`depth/` 256x192 uint16 millimetres; `confidence/` 0, 1, 2 with only 2 used;

`odometry.csv` camera-to-world poses, quaternions x, y, z, w, per-frame

intrinsics scaled by 256/1920 for depth. Unprojection is OpenCV convention.

Getting it wrong is silent, so a regression test pins the floor of c00a170fe1 at world y = -1.48 m within 2 cm (tests/test_stray.py).

Devices. Nokia 8.1 shot every video loop and the bedroom_2_repeat stills; Moto Edge 50 Neo shot the other stills. EXIF make and model were stripped, so attribution is the operator's record and the files say so.

No iPhone was available for capture: every lidar result comes from the three supplied scans, so the lidar capture route is written from the format and has not been walked through on our own device (docs/device_matrix.md).

Runtime on an M1 Max: lidar 4.0, 13.2 and 27.8 s for 1715, 5251 and 9745 frames, against a 3 minute budget. Video is about 8 minutes a clip and photo 77 s for four rooms (docs/benchmark/provenance.json).

Video tier: what was rejected and why

MapAnything was tested and rejected as a metric source

(docs/experiments/mapanything/result.json). Image-only inference on c00a170fe1 at 20, 40 and 80 frames: the median ratio of predicted depth to lidar depth is 0.712, 0.704 and 0.709, standard deviation 0.09 to 0.13. A fixed correction would leave about 13% residual spread, so it is a systematic error with per-frame noise on top, not a calibration.

Feeding it ARKit poses would likely have fixed the scale, and was refused: a video result obtained from lidar-derived poses measures the lidar tier. It is kept for one job it is good at, relative poses across a chunk seam.

What the video tier actually does

COLMAP structure from motion over frames sampled from the clip, run in **chunks** rather than over the whole walk: a 50-second sweep rarely registers as one reconstruction, and a chunk that fails then costs a few seconds of video instead of the room. Chunks are joined by ****floor-plane bridging****, which aligns consecutive chunks on the floor they share and uses MapAnything only for the relative pose across the seam.

Scale comes from the camera-height prior, not from a depth model: each chunk is scaled so its own camera sits 1.40 m above its own floor. Depth Pro runs as a **weak second opinion** and is used to sanity-check that scale, not to set it. The room is then fitted as a ****single room around the camera path****, because segmenting free space out of a partly reconstructed floor either finds nothing or finds a fragment and calls it the room.

Measured: median wall error 32.9% over 16 walls. Section 6 says why that is still far off, and what it is not caused by.

3. Drift handling

ARKit odometry drifts slowly: yaw creeps and floor height wanders over a few minutes of walking, which smears wall faces and breaks the floor into a wedge. cozmo/lidar/drift.py anchors to the building, not the trajectory. The walk is cut into 5-second chunks. Per chunk: the dominant wall azimuth modulo 90 degrees gives a yaw error against the global Manhattan direction, removed by rotating about the chunk's own centroid; the median height of its floor points gives a vertical offset; then its wall points shift along each axis by the median offset to the nearest global face. Corrections beyond configured limits are rejected, because a chunk that looks 20 degrees off usually has too little

wall to estimate from.

The ablation says it barely matters here

(fix_loop/after/bench/runs/*/drift_report.json):

scan	wall thickness off	on	footprint off	on
c00a170fe1	24.2 mm	23.6 mm	55.4 m2	35.8 m2
1a8384c3f6	32.1 mm	35.8 mm	80.0 m2	86.6 m2
c7d28f72c6	32.5 mm	32.6 mm	65.8 m2	64.7 m2

Thickness improves on one scan, worsens on another, unchanged on the third.

The model summary says why: mean yaw error 0.55 to 1.57 degrees, mean height error 3 to 8 mm, mean shift 1.4 to 2.2 cm. **ARKit's poses were already good enough that there was little to correct.** Room overlap does fall consistently. It stays on because it costs one pass and does no harm; that is the whole case for it.

The video tier has its own drift problem and it is not this one: chunks disagree about scale, not orientation, which section 6 covers.

4. Error budget

Every interval is built from named terms, defined in `cozmo/lidar/uncertainty.py`.

Wall length. Statistical: each bounding face's position error is its residual spread over an effective sample size counting 0.25 m patches, never raw points, because points 2 cm apart on one wall are not independent measurements. Systematic: a 1% depth scale bias scaling with length, plus a 1 cm floor. Half width $1.96 * \sqrt{sa^2 + sb^2 + (bias * L)^2}$, floored at 1 cm. On the lidar scans the median interval is 2.4% of length.

Ceiling height. Per 25 cm cell, ceiling mean minus the floor plane beneath, reported as the median. Half width combines cell spread with the depth scale term. **The spread is deliberately not divided by the square root of the cell count**: cells are not independent samples, and sub-millimetre precision from a handheld scan is exactly the confident garbage the grading penalises. Under 15% ceiling coverage no number is invented: a 2.2 to 3.2 m prior with method `prior_no_ceiling_observed` and a warning. Two of three lidar scans take that path; the third measured 3.06 to 3.08 m across rooms. Ceiling height is the weakest measured quantity at every tier: errors run 11.9 to 85.8 cm at the photo tier and 27.2 to 63.3 cm at the video tier, against a 1.5 cm gate (`docs/benchmark/eval.json`).

Areas propagate from edge lengths in quadrature over the two axes.

Partially observed rooms have length and area intervals doubled and say so in the method string. **Damage extent** with depth runs from a contrast estimate to the box projected on the surface; without depth it is `unbounded_no_metric_depth`, 0 to 4 m2, with a warning.

Ground truth, and its limits

The error budget is only as good as what checks it:

- **The supplied lidar scans have no ground truth at all.** Nobody measured that property, so no lidar accuracy number exists or is given.

- **Own rooms: tape, centimetres to the nearest inch.**

`truth_uncertainty_m` is 0.013 and the coverage check widens the truth by

it, so a pipeline is not marked wrong for beating the reference.

● **The hall is scored on ceiling, openings and adjacency only**

(score_walls: false): an open-plan connector with no measurable wall run.

● **Four windows are present_unmeasured:** neither hit nor phantom, and no calibration item.

● **Staged damage is two A4 sheets,** recorded as

extent_m2_upper_bound: 0.0624, a bound on the mark rather than the mark.

5. Calibration analysis

Calibration is scored separately from accuracy, in `cozmo/eval/calibration.py`:

empirical coverage against the nominal 95%, mean interval width as a percentage of value, and a "confident garbage" count, meaning the truth falls outside the interval and that interval is narrower than the median for its quantity type. The median is per quantity because widths in metres and in square metres are not comparable. Broken down by tier and by quantity.

Measured over every capture with tape, from `docs/benchmark/eval.json`:

group	n	coverage	mean width, % of value	confident garbage
all	37	0.97	251	1
photo	16	0.94	43	1
video	21	1.00	409	0
photo, wall length	12	1.00	42	0
photo, ceiling height	4	0.75	44	1
video, wall length	16	1.00	519	0
video, ceiling height	5	1.00	57	0

Both tiers over-cover, and they do it for opposite reasons. The photo tier reaches 0.94 with intervals about 43% of the value: wide, but in proportion to how much a handheld photo reconstruction actually knows. The video tier reaches 1.00 with intervals averaging **409% of the value**, which is not calibration but an interval so wide it cannot be wrong. A wall reported as 6.6 m plus or minus 5 m contains the tape reading and tells nobody anything.

That is the honest reading of the calibration score: the video tier is not being rewarded for knowing its own error, it is being rewarded for saying almost nothing. The one confident-garbage case is a photo-tier ceiling. The lidar tier contributes no coverage, because the supplied scans have no tape. What it does contribute is refusal: two of three scans report ceiling height as a 2.2 to 3.2 m prior rather than a number, and every room in the largest is flagged partially observed with doubled intervals.

6. The fix loops

Two loops were run. Both are negative results and both are reported as such.

Loop 1: cross-capture repeatability (`fix_loop/`)

Declared: worst gate repeatability, 0 of 112 wall rows within tolerance on two lidar captures of one apartment. Root cause hypothesis: room partition is unstable because it grows from observed floor and the walked path, so it follows the operator rather than the building. Predicted after: 40% of rows within tolerance.

The eval was broken first. Plans are emitted in the ARKit world frame, so

the two captures sat 58 degrees apart and no quarter turn aligned them. Every wall pair failed the parallel test. Fixed by canonicalising each plan to its own Manhattan frame before the search, in commits before the BEFORE run.

The falsifier fired. Dropping ghost faces alone, the cheap test named in the declaration, moved nothing: about 1% of face length is rejected and the gate stayed at 0. That also showed the declaration's own evidence had over-counted ghosts at 16%, because it measured faces against an envelope built from the polygons under test.

The fix shipped and did not work. Wall-driven cell complex segmentation with support from the 1.0 to 1.6 m band. Still 0 rows within tolerance, rooms paired across the captures fell from 6 to 3, footprints inflated. Three of four predictions wrong. erosion is the default again.

The cause was coverage, the third falsifier: only **56% and 37%** of each capture's wall-face length has any counterpart in the other. The contrast that carries it, from `fix_loop/evidence/evidence.json`:

measured at	disagreement between the two captures
wall face position	1.9 cm median
polygon edge length	97.5 cm median

The geometry agrees to two centimetres. The partition into rooms and edges does not, and no rule can invent a wall one capture never saw.

Loop 2: video-tier metric scale (`fix_loop/loop2_video_scale/`)

Declared: the video tier's metric scale came from a monocular depth model and was wrong. Eight predictions were written down before any fix code.

Two came true. Every clip now produces a plan, where three of six produced nothing. The share of the sample scan reaching the output went from 30% to 70%, accepted bridges from 2 of 6 to 4 of 6.

Six did not. Median wall error went from **54.8% to 35.3%**, against a predicted 8 to 20%. Nothing reached the 3% gate. The two captures of one bedroom, which should have agreed once both were pinned to the same prior, disagreed **more** than before: 198% and 145% per wall against 107% and 143%.

Three of five falsifiers fired.

Two things the diagnosis missed:

● **The prior only works when the floor plane is right.** A chunk covering a few seconds of sweep that sees almost no floor fits a plane that is not the floor, and the prior then scales confidently to the wrong thing. On one clip two bridged chunks landed at 0.148 and 0.247 metres per SfM unit, a 67% disagreement, both set by the prior. Repeatability got worse because the error stopped being a shared bias and became random, and random errors do not cancel between two captures.

● **The rooms were never measured at all.** A face counted as a wall only if it reached 1.5 m above the floor, a threshold ****inherited** from the lidar tier^{**}. A phone carried at 1.4 m and pointed level or down never reconstructs that high. On one clip all eight faces topped out between 0.72 and 1.46 m, none qualified, and the room came back as the camera path in a box. That was true for every clip in loop 1, and it should have been found by looking at what the room fitter was using before the fix was designed.

Iteration 2 was declared, tried and rejected. Running the clip's frames through the photo tier's reconstruction was predicted to give 3 to 12% median error; at 16 frames it gave +69% and +63%, at 24 frames it gave -36% and -34%. SfM remains the default.

The finding that matters most is not about scale. Nine composed stills of bedroom_2 through the photo path, with the same height prior and the same room fitter, give walls within about 1%. Fifty seconds of video of the same room through SfM gives 53% and 66%. Same scale cue, same fitter, same tape.

The difference is the capture, not the algorithm: nine deliberate stills from the corners beat a continuous walk, because the walk never holds a viewpoint long enough to triangulate a wall well. That is a capture-protocol finding, and it is reflected in docs/capture_protocol.md.

7. Known failure modes

Measured.

- **The video tier is not accurate enough to ship.** 32.9% median wall error, worst 137.4%, and intervals averaging 409% of the value.
- **Cross-capture repeatability fails at every tier.** 0 of 153 rows on the lidar pair, 0 of 8 on the same-device video pair.
- **Ceiling height fails at every tier,** 11.9 to 85.8 cm against a 1.5 cm gate, and is unavailable entirely on two of three lidar scans.
- **Damage precision on photos.** 8 to 15 false regions in a room with 2 marks, from 63 before filtering; on a lidar capture the same filters reach 0 from 106. Two of the four filters need depth and poses.
- **Cracks are the class most at risk:** the verifier that removes false positives removes the crack at threshold 0.20, so 0.15 ships.
- **Room segmentation over-segments:** 11 rooms on a flat with perhaps 6 or 7.

Hazards in the real captures (docs/hazards.md, each with evidence).

Glass and a wardrobe mirror return confident depth for a room that is not there; the damage filters handle them, reconstruction does not. A dog walked through one capture. Glossy tiles thicken the floor peak. Textureless white walls give sparse returns, part of why coverage differs so much between two walks of one flat. An unobserved ceiling is the commonest cause of a wide interval and is purely a protocol problem.

Assumptions that will break. Manhattan: a curved or 45 degree wall appears as a missing face rather than a wrong one, the safer failure but still a failure. Video and photo scale depends on the phone being held near 1.4 m; a clip shot at waist height is wrong by the ratio of heights and nothing in the pipeline can detect it.